A parametric 3D-printed body-powered hand prosthesis based on the four-bar linkage mechanism

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***Abstract*—The widespread of 3D-printing technology has resulted in the appearance of many open-source prosthetic hand models, especially for partial hand amputations. However, most of these designs are not editable and while some are parametric to some degree, customization for every user is limited to scaling the size of a base design. As consequence, most prostheses fail to closely match the user specific anthropometry and have poor aesthetics, which could result in abandonment of the device. Furthermore, achieving a high degree of customization could be a time-consuming task and requires previous knowledge of CAD design. This work presents a prosthetic hand easy to customize by changing parametric dimensions of the finger phalanges and palm on an Excel sheet. Additionally, the design tackles common issues from previous 3D-printed body-powered prosthetic hands by incorporating new features such as the use of linkages instead of cables as finger flexors and a new cable-adjusting system which requires no additional tools and makes the tensioning of finger tendons easier and quicker.**

Keywords---- upper-limb prosthetics; parametric design; 3D-printing; body-powered prosthesis; rehabilitation

I. INTRODUCTION

Hand prostheses are highly customized products due to the variety of amputation levels and the uniqueness of every amputee’s stump. 3D-printing technology has fueled the development of low-cost hand prostheses designs. Nonetheless, these prostheses often fail to match the size and shape of their human counterparts. These models are usually not in editable format and, in consequence, replication and adaptation for a specific user is limited to scaling the size of the prosthesis. Even if the prosthesis’ files can be modified, reshaping a hand design for every user would be a time-consuming task if the model had not been created using geometrical features compatible with dimensional changes within the appropriate ranges of hand athropometry. This issue could be tackled with parametric design of the prosthesis’ parts; that is, with easily modifiable dimensions such as finger phalanx length or breadth, which would allow size customization without the burden of redesigning each component.

Designing the prosthesis parametrically is not a trivial task, especially if the shape of the components is desired to be organic and aesthetic, which is a priority in prosthetics, and if the mechanisms for finger motion are complex.

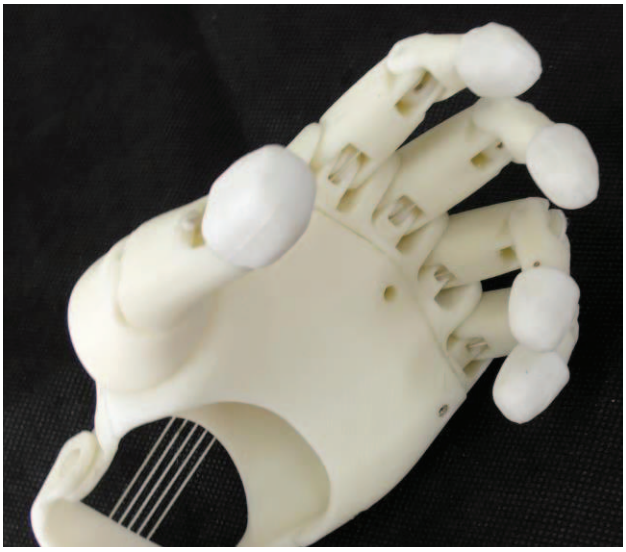

Figure 1. Body-powered hand prosthesis prototype.

Flexible joints [1] or hinge-like joints [2, 3], which are the most common type utilized by open-source body-powered (BP) prostheses, greatly simplify the design of the fingers, which would only consist in three pieces as phalanges coupled by cables or cords. According to Ten Kate et al., every 3D-printed hand prosthesis uses cables or cords as flexors [4]. It is understandable that this is the preferred activation method given the simplicity in assembly that one-piece phalanges joint by tendons provide. Parametric design for length and breadth of the phalanges is also easier to achieve for these kind of finger mechanisms. On the other side, mechanical linkages are used as flexors only for some forearm or transradial-level prostheses, but are more common in powered prostheses. However, adapting the shape of mechanical linkages to achieve an anthropomorphic and aesthetic finger is a difficult task which becomes harder the more joints and coupled links the finger has, as can be noted from previous underactuated finger linkage mechanisms [5, 6, 7]. Moreover, parameterization of the dimensions of linkages with complex shape requires a more intricate set of equations, since every link length must be updated to change the overall length or breadth of one phalanx. Nonetheless, an effort was made to develop an

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anthropomorphic 3D-printed BP hand prosthesis driven by linkage mechanisms which still resembles its human counterpart as can be seen in Fig. 1.

The use of linkage mechanisms can also help to overcome disadvantages of common cable-driven robotic fingers, such as friction, poor trajectory control and frequent loosening of the activation cables. The design process of the parametric model will be explained in the following sections. Additionally, a prototype of a complementary software was developed for the use of prosthetists that allow them to change the dimensions of the parametric segments of the prosthesis without requiring previous knowledge of CAD design.

II. PARAMETRIC DESIGN OF THE HAND

To fully customize the hand with parametric design, first, a standard base design must be created considering each finger length and breadth, as well as the length and breadth of the palm. These different dimensions must be related to each other by proportions or linear functions. Proportions between the finger phalanges related to the total finger length or the hand length have been studied extensively before [8, 9]. Nonetheless, the complex geometry of the palm has not been characterized to a degree in which the 3D modelling task is straightforward and depends heavily of each designer's unique approach. One way to overcome this inconvenient is by 3D-scanning the hands of many subjects to find a suitable set of proportions that can be later transferred to a 3D CAD model. After attaining the design of a suitable parametric CAD model for the hand, creating a model for a specific user requires only to measure the healthy hand and edit the base design. For the parameterization method described in the following subsection, the palm length is the dimension measured from the wrist to the middle finger base and the palm breadth is measured at the level located above the thenar eminence. For the fingers parameterizaton, their total length and the middle finger breadth are measured.

A. Parameterization of the palm

To perform the parametrization of the palm, the dimensions of the hands of three people that fit in the characteristics of the 5th, 50th and 95th human percentile were analyzed. This criterion was based on anthropometric concepts of industrial design applied to a Peruvian population, where the 5th percentile is a woman of 1.5 m in height, and the 95th percentile is a man of 1.8 m in height. Thus, the hands of people who fit these characteristics were 3D scanned.

For each subject, the 3D scan file in STL format was studied on the software Rhinoceros 5.0. Different transversal sections of the hand were measured and contours were drawn using primitive geometric figures, such as ellipses and arcs, to fit the borders of the hand section. The number of sections and their locations along the longitudinal axis were defined through empirical testing in order to obtain the most aesthetic and realistic representation of a human hand. Additionally, the choosing of some sections was intended to enable the

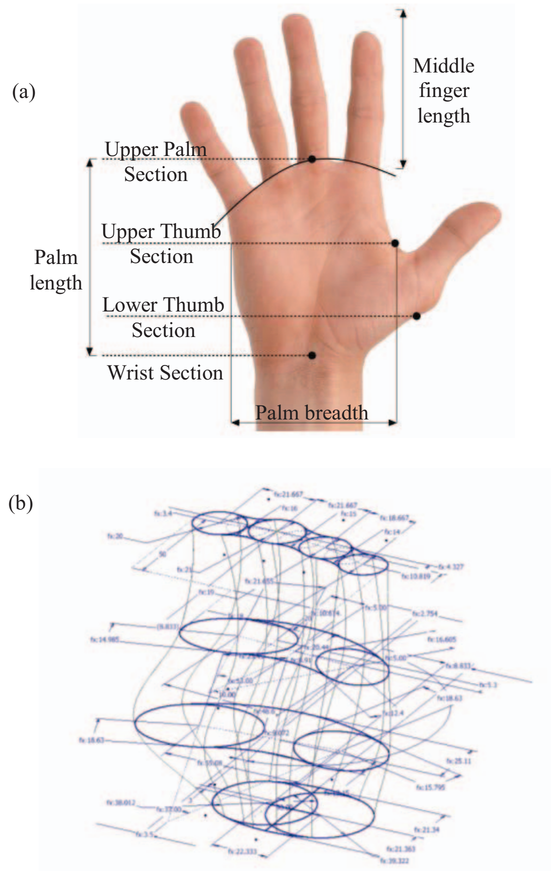
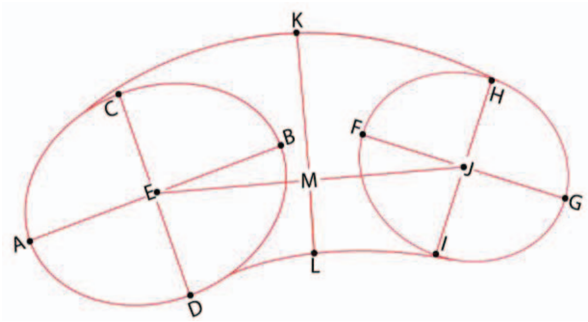


Figure 2. (a) Sections considered for the construction of the palm and (b) their corresponding transversal sketches.

correct positioning and orientation of the thumb in order to perform pinch and tripod grasps.

The four sections selected for the parametric palm were the following: upper palm, located at the level of the middle finger base; upper thumb, located on the upper intersection of the thumb and the palm above the thenar eminence; bottom thumb, located below the thumb's thenar eminence; and wrist, located at the level of the wrist flexion axis (see Fig. 2). Based on the shapes of the contours for each section, two types of sketches were designed: the first one, the 'palm shape' sketch (see Fig. 3), was used for the upper thumb, bottom thumb and wrist sections; and the second, the 'fingers' base' sketch, was used only for the upper palm section (see Fig. 4). The relation between diameters, axes lengths and center distances were studied for designing a palm model that can be easily modified to closely match the user's healthy hand.

To analyze the dimensions of the primitive geometric figures and define the proportions, the magnitudes for each dimension obtained in the analysis of the 3D scans was divided by the corresponding palm breadth for each of the three hand sizes. Then, for each dimension, the three proportions obtained are averaged, obtaining a coefficient which can be used in the parametrization. This process was performed for all values.



A - B: Ellipse A Long Axis
F - G: Ellipse B Long Axis
E - J: Ellipse Center Distance
M - L: Center to Palm
C - D: Ellipse A Short Axis
H - I: Ellipse B Short Axis
M - K: Center to Dorsum

Figure 3. Dimension of of palm shape sketch for the bottom thumb section.

In the palm shape sketch, the most important dimensions are the semi-major and semi-minor axes' lengths of the ellipses, the distance between their centers (EJ segment in Fig 3) and the radial distance from the sketch center to the arcs that define the dorsal (MK segment) and palmar area (ML segment). Table I shows the measurements for each of these variables and for the three percentiles in the case of the bottom thumb section using the aforementioned procedure.

In the fingers' base sketch, the important dimensions are the breadth of the fingers, the distances between the fingers' centers and the plane P (see Fig. 4) parallel to the sagittal plane, which determines the shape of the arc of the hand, and the horizontal distances between these centers (parallel to plane P). For the fingers' breadth, represented by the circumferences, equations derived from the study of the 3D scans were included to simplify the CAD modeling of the fingers. The relation between these dimensions in the hand parameterization is shown in Table II. The index,

TABLE I. RELATIONSHIP BETWEEN DIMENSIONS OF THE BOTTOM THUMB SKETCH AND PALM BREADTH

Variable	Coefficient	Palm Breadth		
		76 mm (5th)	*86 mm (50th)*	*96 mm (95th)*
Ellipse A Long Axis	0.58	44.08	55.68	49.88
Ellipse B Long Axis	0.46	34.96	44.16	39.56
Ellipse A Short Axis	0.46	34.96	44.16	39.56
Ellipse B Short Axis	0.39	29.64	37.44	33.54
Ellipse Center Distance	0.68	51.68	65.28	58.48
Center to Dorsum	0.31	23.56	29.76	26.66
Center to Palm	0.15	11.4	14.4	12.9

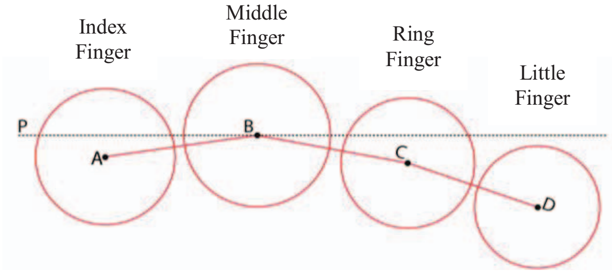


Figure 4. Fingers' base sketch

ring and little finger breadths are calculated subtracting 1 mm, 2 mm and 3 mm to the middle finger breadth respectively, while the thumb breadth is calculated by adding 4 mm. The distances between the centers parallel to plane P are constrained by the palm breadth and by assigning equal spacing between adjacent fingers. Finally, the distances between the fingers' centers and the plane P parallel to the sagittal plane are determined by the condition that when the fingers are fully flexed all the fingertips lie in the same plane parallel to P. In addition, the separation between the planes of the transversal sections mentioned before were also defined comparing the 3D-scans of hands of 5th, 50th and 95th percentiles, but for this case the distances were proportional to the palm length instead of palm breadth.

A previously shown, only the palm length and breadth of the user is needed to obtain a new custom palm design. Fig. 2b shows the drawing of the base palm using these sketches. Additionally, to verify the validity of the parameterization procedure and the values of the proportions involved, palm models were generated using the left hands' dimensions of a total of 25 healthy subjects (10 females and 15 males aged from 19 to 30 years old) and compared with their 3D-scan. Just by visual inspection, the resulting palm models were considered very similar to their human counterparts, as will be shown with the actually implemented prostheses.

Because there are parts of the palm design that inevitably must be manually designed for each user, the palm generation procedure continues in the following manner:

- Using the ratios previously described for the four sections of the palm, a base model is generated with the user's healthy hand dimensions. This base model already includes the supports for the

TABLE II. RELATIONSHIP BETWEEN MIDDLE FINGER WIDTH AND PALM BREADTH

Palm breadth (mm)	*Middle finger breadth (mm)*
70-76	17
77-82	18
83-88	19
89-94	20
95-100	21

fingers proximal phalanges (MCP joint supports).

- Then, the user's residual limb is fitted. The shapes of different users' stumps are very diverse depending on the level and cause of amputation and, for traumatic amputations, the surgery could require the rearranging of the amputee's soft tissues in order to cushion and protect bony prominences, which results in a thicker stump.
- Afterwards, the palm sections thicknesses are modified while keeping the palm breadth constant until the stump fits correctly and enough space is secured for the flexor cables' channels.
- Finally, the channels for the tensioning cables are designed within the palm (see Fig. 5).

B. 3D modelling of the hand

The hand components were designed using the software Autodesk Inventor 2016 and Rhinoceros 5.0 for the prosthetic socket, which is user-specific and has to be designed manually. While some dimensions are fixed, the ones that are parametric are given a specific name and written in equation form. The Autodesk Inventor files (.ipt) are linked to an Excel sheet which contains the measurements taken from the user and generates all the other parameters.

The geometry of the external components was designed as organic as possible within the limitations of the inner mechanisms of each finger. The fingers were designed with a conical shape that resembles the human fingers. For the palm, the aforementioned transversal sections were blended to generate a smooth shape (in Autodesk Inventor, this is called a *loft* operation). Other features for the assembly of the prosthesis, like the channels for the tendons and the support structures for the joints located on the bases of the fingers, were also designed parametrically to adapt for different palm shapes and finger sizes. The hand can be sized even for a 5th percentile female (165 mm hand length [8]) without the limitations of the fabrication process (FDM 3D printing) affecting the quality of the parts. This size is comparable to that of some myoelectric prosthesis designed for female users, like the Bebionic Small [11] and RIC Arm [12].

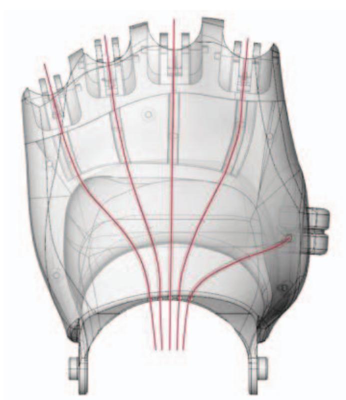


Figure 5. Front view of the generated parametric palm. The lines in red correspond to the activation cable channels.

III. Hand Mechanisms

Each finger was initially designed consisting in three four-bar linkages similar to the mechanism shown in [10]. Each link length was computed according to Grashoff equations and each user's anthropometry (i.e. each finger length). The joints positions are aligned with the finger rotation axes. For example, the MCP joint in the prosthesis is aligned with the knuckle of a human hand. The thumb is made of a single four-bar linkage for the proximal and distal phalanx. The thumb metacarpal is a single part coupled by the flexion cables. Fig. 6a and 6b shows the mechanisms for each type of digit (fingers and thumb), while Fig. 6c shows the hand prosthesis CAD model designed with these mechanisms. Due to the space constraints considered in the 3D modeling of the finger, the mechanisms was simplified by fixing the DIP joint flexion angle at 20 degrees. Thus, for the CAD model only two four-bar linkages were implemented.

Activation or flexion of the fingers is achieved by tensioning cables like in other BP prostheses, with the difference that in this case the cable pulls the crank of the first four-bar linkage, the one corresponding to the proximal phalanx. This motion produces the rotation of the next phalanges by their crank links. The advantage of using these mechanisms is that a considerably smaller wrist flexion angle is needed in comparison with typical hand prostheses to achieve full flexion of the fingers, which would cause less fatigue to the user and would not force them to move far away from anatomical position or make compensatory movements to perform ADL's.

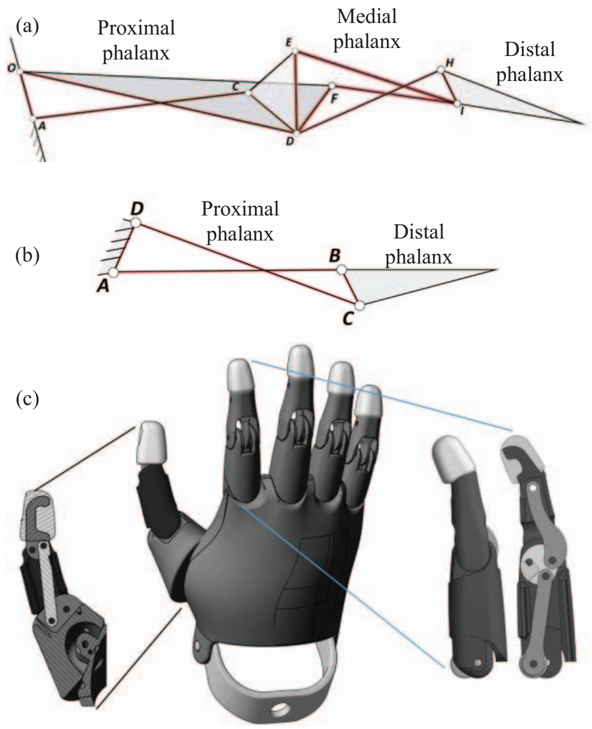


Figure 6. Underactuated mechanisms for the (a) finger flexion and (b) thumb flexion. (c) CAD model of the hand prosthesis and section view of the thumb and index finger.

Regarding the wrist and forearm of the prosthesis, some new features in comparison with common open-source models were introduced. For the wrist, instead of a hinge joint connecting the forearm to enable flexion of the wrist, an additional C-shaped part was added to link these elements. The C-shaped part is assembled to the forearm in a third pivoting point located in the anterior part of the wrist, which adds a degree of freedom corresponding to wrist abduction-adduction (see Fig. 7). This would allow partial hand amputees to perform more complex movements for ADL's like, for example, opening a jar. All the joining shafts were designed as part of the components to be 3D-printed and assembled by press-fit.

The flexion cables go through the palm and the forearm to the tensioning system [13]. To avoid having to include screws for the tensioning of the cables like most open-source hand models, a different system was proposed consisting in pins from which the cables are rolled around. The pins are inserted on a hole and fixed by friction contact to maintain the tension. To change the tension, each pin can be pulled out, rotated to roll or unroll the cable and then inserted again (see Fig. 8). The main advantage of this system is that it eliminates the need for an additional tool (e.g. a screwdriver) to adjust the cable's tension.

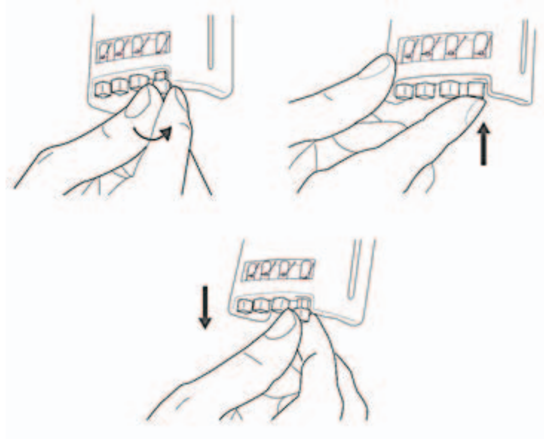


Figure 8. Functioning of the cables adjustment system of the prosthesis.

IV. INTERFACE FOR PROSTHETISTS

A complementary software or add-in for Autodesk Inventor was developed in order to aid the prosthetist in changing parameters and customize the prosthesis in an intuitive manner. With the previous study of the lengths of the hand segments, the parameterization formulas have been included in two parts of the software: an Excel file and a program developed on the Visual Studio development environment using the Autodesk Inventor Application Programming Interface (API). The former is an Excel document which performs the calculations and transfers the dimensions necessary for the design program; the latter receives the calculations and updates the parametric design in Autodesk Inventor through buttons on a user interface before proceeding to exporting the files for 3D printing.

A. Excel spreadsheet

This file is prepares the parameterization data for the add-in (see Fig. 9) and is composed by 4 sheets, each one with a different function. The first is named "For Inventor", in which the parametric dimensions that Autodesk Inventor requires to customize the prosthesis are listed using the same names as in the Inventor files. They are located in the first sheet because this is mandatory to link the Excel file with the Autodesk Inventor files. The second sheet is named "Generate Measurements", which works in conjunction with the third sheet "Calculations" to perform the parameterization calculations in order to obtain the custom model for each user.

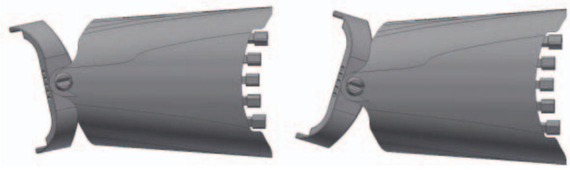
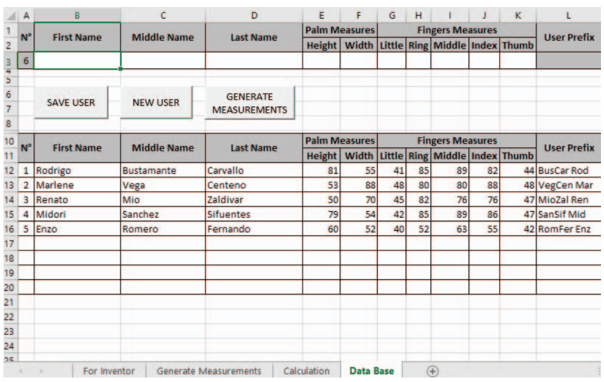


Figure 7. C-shaped link for wrist abduction-adduction.

The final sheet named "Database" is the one that interacts with the prosthetist, where they must enter the lengths of each finger, along with the breadth and height of the palm. Fig. 9 shows the Database sheet, in which the prosthetist enters the name of the patient. The user prefix for the Autodesk Inventor files and date are generated automatically with the patient's first and last names. The interaction buttons are used to save user data, create new users and generate measurements. Whenever required, new user data can be placed and then, by pressing the button "Generate measurements" and saving Excel file, the add-in will be ready to update the measurements in the Autodesk Inventor file. Therefore, the first three sheets are the internal calculations that should only be modified by the programmer or designer but not the prosthetist, who should only be allowed to access to the Database sheet in order to place the user's data.

B. API Inventor Add-in

After generating and saving the user's measurements in



N°	First Name	Middle Name	Last Name	Palm Measures		Fingers Measures					User Prefix
				Height	Width	Little	Ring	Middle	Index	Thumb	
6											

SAVE USER | NEW USER | GENERATE MEASUREMENTS

N°	First Name	Middle Name	Last Name	Palm Measures		Fingers Measures					User Prefix
				Height	Width	Little	Ring	Middle	Index	Thumb	
1	Rodrigo	Bustamante	Carvallo	81	55	41	85	89	82	44	BusCar Rod
2	Marlene	Vega	Centeno	53	88	48	80	80	88	48	VegCen Mar
3	Renato	Mio	Zaldivar	50	70	45	82	76	76	47	MioZal Ren
4	Midori	Sanchez	Sifuentes	79	54	42	85	89	86	47	SanSif Mid
5	Enzo	Romero	Fernando	60	52	40	52	63	55	42	RomFer Enz

For Inventor | Generate Measurements | Calculation | Data Base

Figure 9. Database with its four components (For Inventor, Generate Measurements, Calculation and Database).

the Excel spreadsheet, the prosthetist should use the add-in (see Fig. 10) or interface to modify the Autodesk Inventor files. Modifications can be made directly if needed before exporting the files for 3D printing.

The add-in performs the update of the parametric model by pressing the “Update Measurements” button, thus interfacing between the Database in Excel and the 3D CAD model in order to generate the prosthesis with the user's measurements identified by the name within the textbox “User Prefix”. Another function of the add-in is the possibility to save the files of the hand separately (each part file of the assembly) with the “Save Assembly and Parts” button. It also has been implemented the function to rotate the parts of the prosthesis to a specific angle before exporting them to STL files. This option was added to optimize the orientation of the parts depending on their loading during use because of the well-known weakness of 3D-printed components in their building direction.

V. Implementation and Evaluation

The design described previously was adapted for three different sizes, one medium sized corresponding to a not amputated volunteer, one small sized for a transmetacarpal amputee and one large sized for a congenital amputee at wrist disarticulation level. For the users with amputation, both the stump and healthy hand were scanned using a Sense 3D Scanner to design the socket and adjust to the size of every element according to the user’s anthropometry. Additional measurements of the stumps and healthy hands were taken to verify the precision of the 3D-scan dimensions and scale the model if necessary. Additionally, the medium size prosthesis was created to verify the adaptability of the parametric model to different sizes. Note that the medium size prosthesis has a greater total length because of finger lengths, but its palm is smaller than the large one (see Fig. 11).

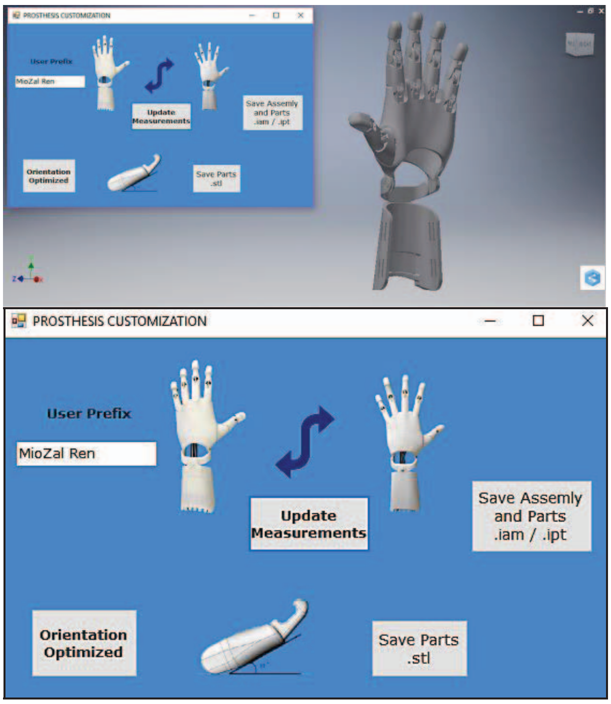


Figure 10. User interface (add-in) for Autodesk Inventor.

The prostheses were manufactured using ABS plastic and then assembled. For the small prosthesis, the resulting model closely matched the user’s healthy hand, as can be seen in Fig. 9a. The prosthesis for this user, who had a hand length of 165mm, had a weight of 200 g considering the Velcro® belts and silicon pads attached to the inner surface of the socket, which is similar to the weight of a Cyborg Beast prosthesis for a 16 year-old child (184.2 g [2]). The cost of the materials for this prosthesis was approximately $80. However, the complexity of the finger causes an increased time in assembly which could make the manufacturing more expensive. This time can be in some way compensated with the easier cable routing, since the cables only go through the palm and forearm cover.

The donning and doffing of the prosthesis was taught to the user, who learned quickly and was comfortable with the force required to close the fingers. The small wrist flexion required also influenced on the ease of use. The flexion angle was related to the small cable excursion required to completely close the fingers, which was about 4.5 mm, a remarkably small value compared to the typical

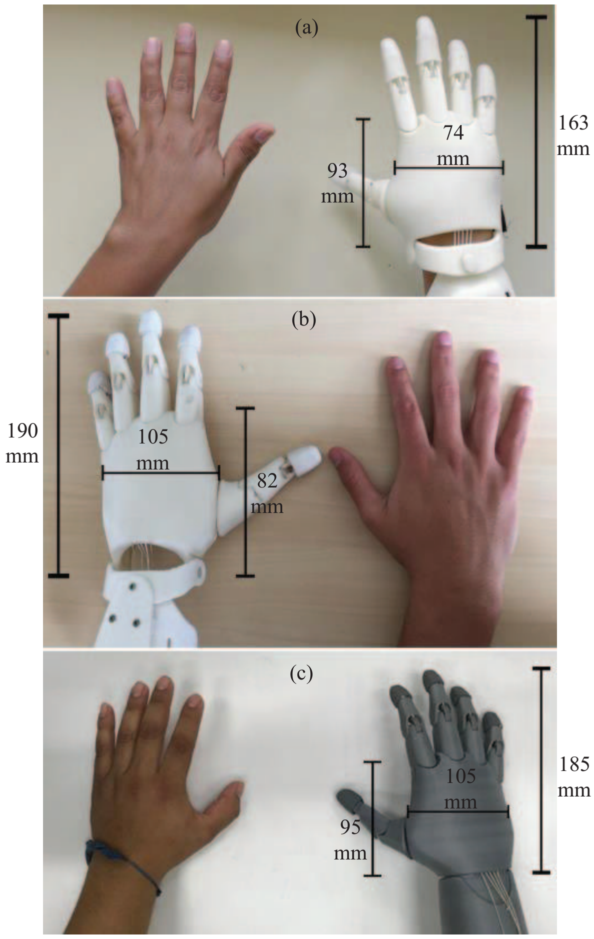


Figure 11. Comparison of human hands and 3D-printed prosthetic hand for users (a) small, (b) medium and (c) large sizes.

cable excursion in BP prosthesis (37-60 mm [14]).

After mastering the activation motion, the user was asked to move objects from different shapes and sizes from one place to another on top of a table. No more than 5 trials were needed to succeed for every object. The most remarkable fact was the ability to effortlessly carry objects with smooth surfaces thanks to the silicon rubber finger covers. The smallest object carried during this first trial session was a 20 mm diameter sphere (see Fig. 12). Moreover, by the fourth training session with the prosthesis, the user could already perform more complex tasks such as building a tower of blocks or picking up and placing buttons or coins (see Fig. 13).

VI. Conclusions and Future Work

A prototype of a new type of body-powered parametric hand prosthesis was presented. It was shown that it is possible to customize the original design for different hand sizes with proper variations of the dimensions of each segment instead of a volumetric size scaling. Despite the number of links per finger, the resulting model is aesthetic and functional. The small wrist flexion angle needed to activate the fingers makes it easy to use and comfortable.

The trajectory is determined by the finger mechanism. Moreover, this trajectory mimics the human fingers and is appropriate to grab a wide variety of small and large objects. However, a better grasping force could be attained with fingers adaptability, which would allow force distribution (when some fingers are halted by an object, the others can continue their motion). In our case, the four bar linkages only allow equal force distribution.

Our complementary software is on a very early stage and more features and usability upgrades should be added to make it intuitive for prosthetists. A next step would be integrating the Excel file and the add-in in a single program. Furthermore, the design itself has room for improvements, such as the adding of a silicone glove cover for the whole hand and parameterization of the socket geometry for easy adaptation to different shapes of stumps.

Figure 12. Different objects grasped by the prosthesis.

Figure 13. Snapshots of a complex activity performed with the use of the prosthesis: picking up coins or buttons.

Standardized functionality test are pending to perform objective comparisons with other existing BP prostheses. Finally, force and durability mechanical testing of the prosthesis is a matter of further study to upgrade this prototype model to a commercial product.

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